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ODROID-N2

N2 PCB revision and LOT number history

LOT(Year+Month)	PCB Revision	Description
1901	rev0.3 20190117	Sample
1903	rev0.4 20190307	1st Mass Production
2006	rev0.5 20200406	N2+
2010	rev0.5 20200406	As the ODROID-C4 board, the 3.3V(Pin#1) power rail of the 40pin expansion connector has been changed to be Always-ON in the shutdown mode.
2011		
2102	rev0.5 20200406	To lower the electrical stress to the GL3523 hub controller, the GL3523 power source has changed to 5V from 3.3V by using internal regulator. Mount a zero ohm resistor between the source and drain of Q1 instead of the FET As a negative side effect, the PWR red LED does dimly turn on even when the system turns off. This issue appears with the production batch of EL2102 only.
2106	rev0.6 20210121	The power of GL3523 is completely isolated from the 3.3 V system power rail. Between the VDDOAO_3V3 and the VCC3V3 net, it was changed to integrated load SW instead of discrete PMOS FET.

2111	rev0.6 20210121	The SPI flash memory manufacturer has been changed to XTX from MXIC, and the firmware version of Petiboot been updated to 20211112 to work with the XTX flash memory.
2202	rev0.6 20210121	Due to the semiconductor shortage, the maximum DC input voltage was lowered from 18V to 16V to use to FirstSilicon's FC6811 from FC3470.
2208	rev0.6 20220517	the rev0.6 20210121 version and schematic are the same, but the footprint of the low acoustic capacitor on the PCB has been changed.

We did not change the PCB revision because any key features were not changed nor added.

How to check year and month with the production LOT number

The label near the eMMC connector shows the year and the month of production in four digits in square red as shown below.



Schematics, Drawings and S922X datasheet

N2+	N2
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* N2 Schematics & PCB Mechanical drawings (DXF CAD format)

(<https://dn.odroid.com/S922X/ODROID-N2/Schematic/>)

* Amlogic S922X Data Sheet (https://dn.odroid.com/S922X/ODROID-N2/Datasheet/S922X_Public_Datasheet_V0.2.pdf)

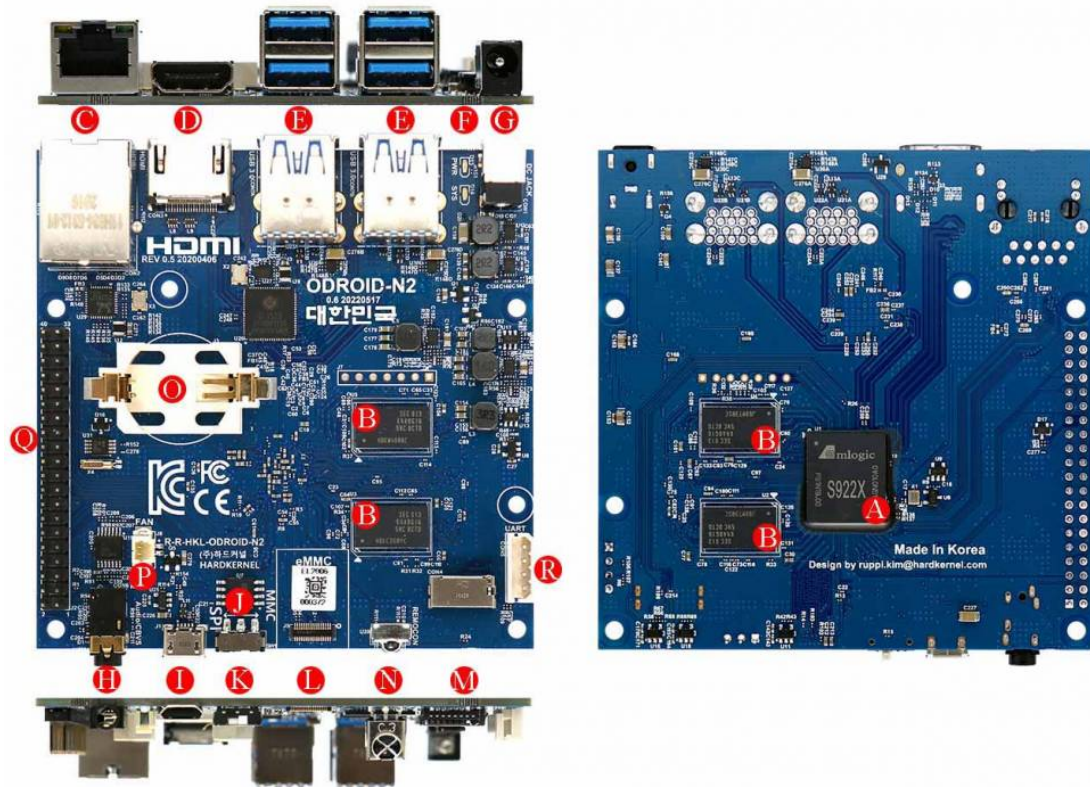
* ODROID-N2+ Heatsink (STL format)

* ODROID-N2+ Heatsink (DXF CAD format)

* ODROID-N2+ Fritzing part(Breadboard view)

Board Layout

N2+	N2
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Specifications

	N2	N2+
Form Factor	Board Dimensions: 90mm x 90mm x 17mm Heatsink Dimensions: 100mm x 91mm x 25mm Weight: 190g including heatsink	Heatsink Dimensions: 100mm x 91mm x 18.75mm Weight: 200g including heatsink
Processor	Amlogic S922X Processor (12nm) Quad-core Cortex-A73(1.8Ghz) and Dual-core Cortex-A53(1.9Ghz) ARMv8-A architecture with Neon and Crypto extensions Mali-G52 GPU with 6 x Execution Engines (800Mhz)	Quad-core Cortex-A73(up to 2.4Ghz) and Dual-core Cortex-A53 (up to 2Ghz)
Memory	DDR4 4GiB or 2GiB with 32-bit bus width Data rate: 2640 MT/s (PC4-21333 grade) 1.2Volt low power design	
Storage	0	

	N2	N2+
Networking	1 x GbE LAN port (RJ45, supports 10/100/1000 Mbps) - Realtek RTL8211F (Ethernet transceiver) - LED indicators * Green LED: Flashing by data traffics at 100Mbps connection * Amber LED: Flashing by data traffics at 1000Mbps connection You can use an optional WiFi/Bluetooth USB adapter. There is no wireless connectivity on the board.	
Video	1 x HDMI 2.0 (up to 4K@60Hz with HDR, CEC, EDID) - Although the +5V Power pin is able to supply a minimum of 300mA, according to HDMI 2.0 specification, a sink shall not draw more than 50mA of current from the +5V Power pin. 1 x Composite video (3.5mm jack)	
Audio	1 x Stereo Audio line-out (3.5mm jack) 1 x HDMI digital output 1 x Optional SPDIF optical output	
External I/O	4 x USB 3.0 Host ports (shares one single root hub) - the current limit of the USB Load Switches are typical 2.4A 1 x USB 2.0 OTG port for Host or Device mode. (No power input) 1 x Debug serial console (UART) 1 x Peripheral Expansion Header (40-pin, 2.54mm pitch) - 2 x DC 5V, 2 x DC 3.3V, 1 x DC 1.8V, 8 x GND - 1 x SPI - 2 x UART - 2 x I2C - 25 x GPIO (Max) - 2 x ADC input (10bit, 1.8V Max) - All 3.3V I/O signal level except for ADC input at max 1.8Volt.	

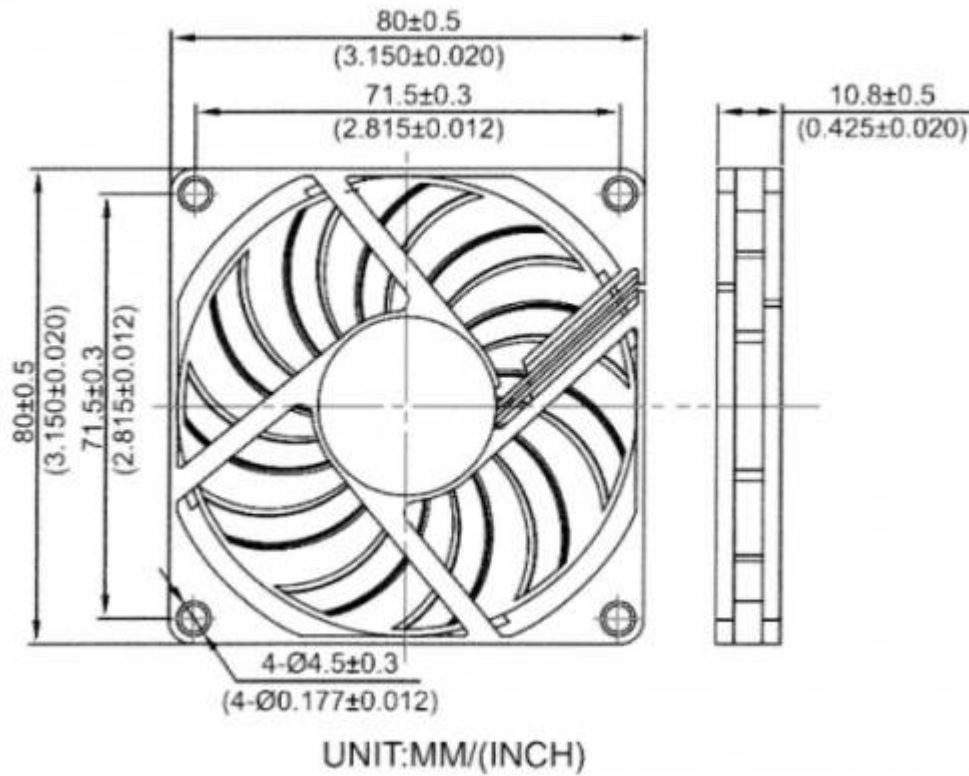
	N2	N2+
Other features	On board RTC(Real Time Clock) IC to keep date and time 8MiB SPI Flash for future BIOS implementation IR receiver for remote controller Built with a large passive heatsink System LEDS Indicators: - Red (PWR) – Solid light when DC power is connected - Blue (ALIVE) – Flashing like heartbeat while Kernel is running. Active Cooling Fan Connector (5V 2-pin) - Connector (2-pin, 1.25mm pitch)	A CR2032 coin battery holder was replaced with the 2-pin 1.25mm pitch connector. No battery is included in the board. Please purchase the very common CR2032 from your local store. You can keep the RTC time up to several years with a 200mAh CR2032 battery.
Power	1 x DC jack : outer (ground) diameter 5.5mm, inner(positive) diameter 2.1mm DC 7.5V ~ 18V (up to 25W) - DC 12V/2A power adapter is recommended Power consumption: - IDLE : $\approx$ 1.9W (Performance governor) - CPU Stress : $\approx$ 5.5W - Power-off : $\approx$ 0.2W	1 x DC jack : outer (ground) diameter 5.5mm, inner(positive) diameter 2.1mm DC 7.5V ~ 18V (up to 25W) 7.5V ~ 16V from 2022 production (LOT number starts with EL22) - DC 12V/2A power adapter is recommended Power consumption: - IDLE : $\approx$ 2.2 ~ 3.3W (Performance governor) - CPU Stress : $\approx$ 5.9W (@2016/2208MHz), 6.2W (@2016/2400MHz) - Power-off : $\approx$ 0.2 ~ 0.35W

We recommend powering the ODROID-N2/C4 with a good quality 12V/2A power adapter. However, using a good quality 15V/4A power adapter is also OK if you already have one handy.

Note that 15V/4V is not necessary for the N2, N2+ and C4, 12V/2A is plenty enough. On the other hand 15V/4A is required for the HC4 and the H2 series because these boards may power mechanical hard drives.

FAN

This FAN is only compatible with ODROID-N2+ Heatsink



1. Rated Current : 0.18A/5V/1,600 RPM
2. Air Flow(CFM) : 11.2
3. Air Pressure(In mmHO) : 0.07
4. Noise Level (dBA) : 19.6
5. Bearing : Sleeve
6. Cable length : 100mm

Connectors

Expansion Connectors (J2)

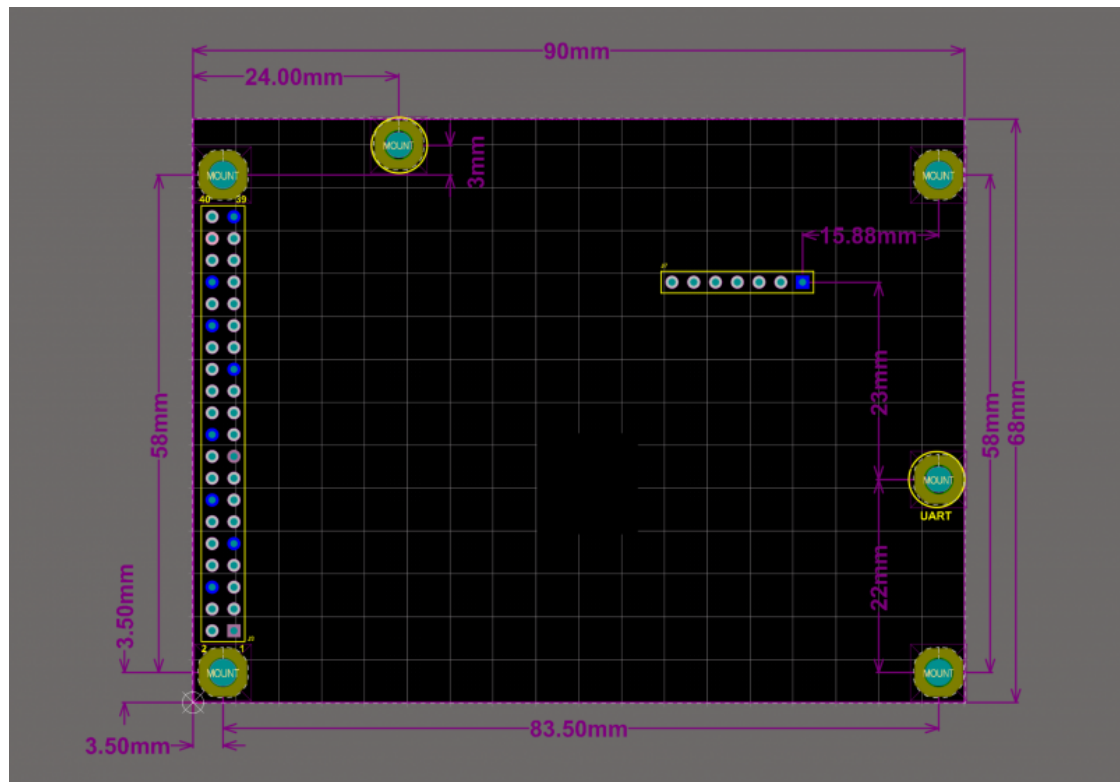
Expansion Connector Description

UART Console Connector

UART	
Pin 4 -	GND
Pin 3 -	RXD
Pin 2 -	TXD
Pin 1 -	VCC

CON5
3.3V LVTTL

Add-on or Shield board design template



DWG CAD format: n2_main_rev0.5_20200406_template.zip

Regulatory Compliance Documents

- *ODROID-N2 KC Certification
- *ODROID-N2 CE Certification
- *ODROID-N2 FCC Certification
- *ODROID-N2 RoHS Certification
- * ODROID-N2 RoHS Test Report



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